

Software engineers build the technological infrastructure for our increasingly connected world. They are in high demand to solve complex engineering problems across industries and government, working in areas such as:

- aerospace and space exploration
- automation and connectivity
- biomedical modeling and devices
- data, computing and networking
- human and intelligent systems
- natural and fabricated environments

With an interdisciplinary curriculum that covers large-scale product development and incorporates intensive real-world design, the University of Arizona BS in software engineering prepares students for some of the most coveted technology jobs in the world.

Academic Focus Areas

Covering the Entire Software Development Lifecycle

Bolstered by intensive design experiences that reflect industry demand and focused on large-scale product and system development to meet societal needs, the University of Arizona BS in software engineering goes far beyond programming fundamentals and theoretical applications. The degree program – administered by the department of Systems & Industrial Engineering – is offered on the main campus in Tucson, **Arizona Online**, and **Yuma Campus**. Courses cover the entire software development lifecycle, or SDLC. Among them are:

- Software requirements analysis and testing
- Software architecture and design
- Software assurance
- Project management
- Cost estimation
- Digital logic
- Computer organization

Students choose electives – such as web and mobile applications, or embedded systems – from across many of the college’s eight departments.

Importantly, students learn about software DevOps, Agile methodology, and continuous integration and deployment. They use open-source integrated development environments,

operating systems and cloud-based infrastructures to gain the specialized software skills in highest demand by top employers.

Research Opportunities

Globally recognized faculty and students collaborate across the campus and around the world to tackle challenges in health care, cybersecurity, big data and transportation.

Ongoing research involving software engineering includes:

- Autonomous systems and robotics
- Communications, coding and information theory
- Computer architecture and cloud/distributed computing
- Data analytics, informatics and machine learning
- Embedded systems
- Wireless networking, security and systems

Outside the Classroom

Real-World Learning

Software engineering students do design projects every semester. They get experience participating in clubs and design-build competitions as well as internships with longtime industry partners.

Additionally, valuable contacts are made in student chapters of professional organizations such as IEEE and IISE/INCOSE and at specially scheduled events with industry leaders.

Career Paths

Endless Possibilities

Software engineering graduates work in corporate and government settings as well as in research and as independent consultants. They become software and application architects and developers, project managers, network engineers, and business owners. Software engineers are in high demand across industries with employers such as Google, Amazon,

Microsoft, Snap Inc., Medtronic, Johnson & Johnson, Raytheon, General Dynamics, Honeywell and Wells Fargo. The list, at home and abroad, is limitless.

Build Global Solutions

Design, develop and maintain complex, high-impact software systems for any field.

Software Engineering (SFWE) at the U of A prepares students to build the programs that run our world. Take on every stage of software production – from theory and ideation to lifecycle development, DevOps, agile methodologies, continuous integration and deployment – to create a better digital world.

Curriculum Shaped by Industry

Since its launch in 2021, the program has been guided by top industry partners to meet modern demands. Courses focus on large-scale software product development and system-level design, giving graduates the tools to create solutions that scale. Classes are available on the main Tucson campus, **Yuma**, **Chandler**, or fully online. Degrees are offered for **BS**, **MS**, and **PhD**, including an accelerated master's program.

<https://nearyou.arizona.edu/yuma>

<https://nearyou.arizona.edu/chandler>

<https://ece.engineering.arizona.edu/undergrad-programs/degrees>

<https://ece.engineering.arizona.edu/grad-programs/campus-degrees>

<https://ece.engineering.arizona.edu/grad-programs/campus-degrees>

Hands-On Experience for Job Success

Graduates are equipped for success in virtually any area, including:

- Aerospace
- Artificial intelligence
- Health care technology
- Smart infrastructure
- Financial systems
- Data science

- Automation

High-Demand Career Skills

Graduates are ready to jump into rewarding jobs with companies like Google, Raytheon, Intel, Microsoft, Northrop Grumman, Snap Inc. Honeywell and Amazon, along with national research centers such as Jet Propulsion Laboratory, Sandia and Oak Ridge.

These partners serve as program mentors, ensuring students have experience in the latest tools, cloud infrastructures, and open-source platforms.

<https://grad.arizona.edu/admissions-guides/#program/SFEPHD>

Learn More About This Program

View **Software Engineering PhD** in the University's Main Catalog.

contactsAcademic Program Contacts

Admissions Contact(s)

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Graduate Program Coordinator(s)

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homeDepartment(s) / Academic Units(s)

Electrical & Computer Engineering

domainCollege(s)

College of Engineering

local_atmTuition and Fees

Please refer to the UA Bursar's Office [Tuition and Fees Calculator](#) for up-to-date information about tuition and fees.

Please refer to the UA Registrar's Office [Special Course Fees](#) for up-to-date information about special course fees.

Campus(es) Where Offered

- University of Arizona - Main: Tucson
- Arizona Online: Online

eventAdmissions Deadlines

Domestic Applicants:

- Fall:
- Spring:

International Applicants:

- Fall:
- Spring:

ruleAdmissions Requirements

To be considered for admission to the Doctoral program, the candidate must have attained a Bachelor of Science degree in a computing-related field. Other factors that will be considered are as follows:

- GPA in previous undergraduate/graduate work of at least 3.0
- A minimum of 3 Letters of recommendation
- Proficiency in 1+ programming language OR 1-2 years of industry experience in a software-related position.
- International students should meet the minimum [English Proficiency scores set by the Office of Graduate Admissions](#)
- A match between faculty and student interests

quizStandardized Tests

GRE is now optional for PhD Admissions

If providing scores, the below are considered competitive:

- *Minimum GRE Verbal: 153 (500)*

- *Minimum GRE Quantitative: 159 (750)*
- *Minimum GRE Written: 4.0*

International Applicants:

Proof of English Proficiency is required per the [Graduate College's Guidelines](#) whose first language is not English.

The Graduate College does not accept Duolingo scores for demonstration of English Proficiency.

Acceptable English Proficiency Credentials:

- [Test of English as a Foreign Language\(link is external\)](#) (TOEFL) - minimum score of 79 iBT (or 60 on the revised PBT with no section score lower than 15). Individual MyBest scores must also be dated within 2 years of the enrollment term to be considered valid
 - The TOEFL Essentials Test is not an accepted test of English Proficiency for doctoral programs
- [International English Language Testing System\(link is external\)](#) (IELTS) - minimum composite score of 7, with no subject area below 6
- [Pearson PTE Academic\(link is external\)](#) - minimum score of 60
- Graduate [English Language Endorsement\(link is external\)](#) from the Center for English as a Second Language (CESL)
- [CEPT Full Academic test \(link is external\)](#)- offered by our Center for English as a Second Language (CESL), minimum Total score of 110

paidFunding Opportunities

It is recommended that both international and domestic applicants apply by December 1st if they wish to be considered for priority funding.

The Software Engineering program offers Graduate Assistantships (Teaching & Research) to students demonstrating outstanding performance in their undergraduate and graduate careers.

Unfortunately, we cannot offer Graduate Assistantships to all new incoming students.

check_circle_outlineInternational Conditional Admission

International applicants will be considered for [conditional admission](#) by this program.

dynamic_feedOther Information

The GRE Institution Code for The University of Arizona is 4832.

ETS Major Field Code(s) for this program: Software Engineering

Program-Level Information

Application Acceptance Rate:	33.33%
Avg. Time-to-Degree (years):	

Department-Level Information

Enrollment Percent Male:	82.35%
Enrollment Percent Female:	17.65%
Enrollment Percent International:	35.29%
Enrollment Percent URM:	11.76%

infoAbout these statistics:

- All application, admit, and enrollment data are from Fall 2025.
- Average Time to Degree calculates how long graduates in the 5-year period from 2020-2024 took to complete their degrees, counting back to the first semester entering their program.
- Underrepresented Minorities (URM) includes domestic students of ethnic groups traditionally underrepresented in higher education: African Americans, Hispanics, Native Americans, and Native Hawaiians/Pacific Islanders.

B.S. in Software Engineering Catalog Year 2025-26 Below is the advised sequence of courses for this degree program on Main Campus as of 1/16/2025. Official degree requirements and course prerequisites are in the University General Catalog; prerequisites are subject to change.

Course Number and Title Units Prerequisites/Enrollment Requirements

1st Semester MATH 122A/B or MATH 125 Calculus I with Applications 3/5 Appropriate Math Placement *CHEM 151 Chemical Thinking I or CHEM 161/163 4 Appropriate Math Placement ENGL 101 or 107 or 109H First-Year Composition 3 ENGR102A/B Introduction to Engineering or ENGR 102 3 ENGR102A & 102B: Pre- or Co-requisite of MATH 112 or higher; First-Year Status; College of Engineering Major; must not have taken ENGR102 UNIV 101 Intro to the General Education Experience 1 Semester Total 14/16

2nd Semester MATH 129 Calculus II 3 MATH 122B or 125 with C or higher SFWE 101 Introduction to Software Engineering 3 MATH 112 or higher with C or higher ECE 101 Programming 1 4 MATH 112, 120R, 122B or 125 or placement at the level of MATH 120R ENGL 102 or 108 First-Year Composition 3 ENGL 101 or ENGL 107 *CHEM 152 Chemical Thinking II or CHEM 162/164 or MSE 110 Solid State Chemistry or MCB 181R/L Intro Biology I 4 CHEM 152 and MSE 110: CHEM 151 or 141/143 or 161/163; MCB 181R/L: Appropriate Math Placement level Semester Total 17

3rd Semester MATH 243 Discrete Mathematics in Computer Science 3 MATH 122B, 125 or 129 (transfer credit allowed) *PHYS 141 Introductory Mechanics or PHYS 161H 4 MATH 122B, 125 or Appropriate Math Placement Level ECE 201 Programming 2 3 ECE 101 with C or better General Education: Exploring Perspectives (Artist) 3 General Education: Exploring Perspectives (Humanist) 3 Semester Total 16

4th Semester SIE 277 Object-Oriented Modeling and Design 3 ECE 101 or CSC 110 ECE 274A Digital Logic 4 ECE 101; Concurrent enrollment or completion of MATH 129 SFWE 201 Software Engineering Sophomore Colloquium 1 SFWE 101 or ECE 101 PHYS 241 Introductory Electricity and Magnetism or PHYS 261H 4 PHYS 241 or 261H: PHYS 141 or 140 or 161H; MATH 129 or Appropriate Math Placement MATH/STAT Elective - Consult with faculty advisor 3 Appropriate Math Placement Semester Total 15

*Each of the following foundational science courses satisfies the requirements for General Education: Exploring Perspectives (Natural Scientist): CHEM 151 or 152 or 161 or 162; or PHYS 141 or 161H. Advanced Standing is required for 300- and 400-level engineering courses (see your academic advisor for details).

Course Number and Title Units Comments

5th Semester SIE 305 Introduction to Engineering Probability and Statistics 3 SFWE 301 Software Requirements Analysis and Test (Fall Only) 3 SFWE 303 Data Persistence 3 ECE 369A Fundamentals of Computer Organization (Fall Only) 4 Semester Total 13

6th Semester SFWE 302 or 405 Software Architecture and Design (Spring Only) 3 SFWE 401 Software Assurance and Security (Spring Only) 3 ECE 311 Engineering Ethics and Contemporary Issues 1 CSC 355 Discrete Structures and Basic Algorithms or CSC 345 Analysis of Discrete Structure 3 Technical Elective 3 Consult with faculty advisor † General Education: Building Connections 3 Semester Total 16

7th Semester SFWE 403 Software Project Management (Fall Only) 3 ENGR 498A Interdisciplinary Capstone 3 Senior Status General Education: Exploring Perspectives (Social Scientist) 3 Technical Elective 3 Consult with faculty advisor † General Education: Building Connections 3 Semester Total 15

8th Semester SFWE 402 DevOps (Spring Only) 3 Technical Elective 1 - 3 Consult faculty advisor for course approval ENGR 498B Interdisciplinary Capstone 3 Senior Status Technical Elective 3 Consult with faculty advisor † General Education: Building Connections 3 UNIV 301 General Education Portfolio 1 Semester Total 14/16 † Students should work closely with their academic advisor to select

General Education: Building Connections courses; some course work in the major, such as some Technical Elective courses, may also fulfill General Education: Building Connections requirements.

Overcoming Tech Infrastructure Challenges Software engineers solve complex problems in space exploration, automation and connectivity, and human and intelligent systems. Bolstered by intensive design experiences and focused on large-scale product and system development to meet societal needs, the University of Arizona's curriculum goes far beyond programming fundamentals and theoretical applications. In Step with Industry Students in this interdisciplinary degree program gain specialized software skills in high demand by top employers. Students learn about DevOps, agile methodology, and continuous integration and deployment. Each semester, they develop leadership skills through real-world projects emphasizing teamwork, critical thinking and professionalism. Burgeoning Field U.S. News & World Report ranks software developer, or engineer, as the best technology job in the United States, and the second-best job across all categories. Jobs in the field are expected to grow quickly, and the median salary is over \$110,000, according to the Bureau of Labor Statistics.

ece.engineering.arizona.edu SOFTWARE ENGINEERING Companies want to be part of this because they know they're going to reap the benefits from having highly skilled software engineering graduates." Sharon O'Neal, professor of practice, electrical & computer engineering ece.engineering.arizona.edu Learning From Experience Outside the classroom, students participate in a variety of activities to build leadership skills and prepare for the workforce. • Paid internships with longtime industry partners • Formal networking opportunities with faculty, alumni and industry • Senior design projects with experienced industry mentors • Research opportunities and field experience • Student chapters of professional organizations • National competitions and student clubs, such as Software Engineering Wildcats A Place for Everyone Various engineering clubs – American Indian Science & Engineering Society; National Society of Black Engineers; Out in Science, Technology, Engineering, and Mathematics; Society of Hispanic Professional Engineers, and Society of Women Engineers, for example – help ensure all students feel welcome and connected. Excellence in Education & Research Globally recognized faculty and students collaborate across campus and around the world to tackle some of humanity's biggest challenges. • Autonomous systems and robotics • Communications, coding and information theory • Computer architecture and cloud/distributed computing • Cybersecurity • Data analytics, informatics and machine learning • Embedded systems • Health care systems • Wireless networking, security and systems 520.621.6032 – enr-admissions@arizona.edu 520.621.6171 – undergradadvisor@ece.arizona.edu Recruiting and Admissions Advising Software engineering opened doors I never thought existed. Its diverse skill set caught employers' attention, leading to opportunities and job offers across industries." Amber Parker, student " " engineering.arizona.edu/virtual-tour

Helping You Achieve Your Educational Goals

In addition to the University's **scholarship package** and financial aid options, the College of Engineering provides opportunities for merit- and need-based undergraduate scholarships to offset tuition, fees and other student expenses. Thanks largely to the support of alumni and donors, the college awarded over \$1 million in scholarships this year.

Incoming Freshman Scholarships >

<https://engineering.arizona.edu/undergrad/scholarships#incoming>

Continuing Student Scholarships

><https://engineering.arizona.edu/undergrad/scholarships#continuing>

Free for AZ Native Americans

><https://engineering.arizona.edu/undergrad/scholarships#nativeamerican>

Incoming Transfer Scholarships

><https://engineering.arizona.edu/undergrad/scholarships#transfers>

Scholarship Universe

The **UA Office of Scholarships and Financial Aid** administers most College of Engineering scholarships through **Scholarship Universe**, a free searchable database, which – based on a student-created profile – matches academic, personal, and professional qualities and interests with potential funding. Scholarships are awarded through federal and state agencies, private corporations, professional membership organizations, alumni and donors.

Scholarships for Incoming First-Year Students

New #ENGR Cats may also apply for a wide range of scholarships – using a single, convenient application platform – through Scholarship Universe, the UA's scholarship-matching tool for all admitted students.

Students who apply and are admitted to the UA Honors College may also be eligible for **Honors Scholarships**.

For need-based aid or scholarships, a **Free Application for Student Aid** (FAFSA) must be on file with the UA Office of Scholarships and Financial Aid.

Incoming first-year engineering students may qualify or apply for the following College of Engineering awards:

Engineering Scholar Awards

Thomas R. Brown Distinguished Scholarships



Scholarships for Continuing Students

Current students may complete the yearly General Engineering Scholarship Application in Scholarship Universe to be considered for 200+ departmental and college-wide scholarships with a single form. Application guidelines and deadlines are usually released in February and the process closes mid-March. Be sure to check in with your academic advisor for questions on scholarship timelines.

Want to get a head start on your application? Visit **UA Career Services** for help with your résumé and **UA Writing Skills Improvement Program** for help with your bio.

Remember: For need-based scholarship consideration, continuing students are required to have a **Free Application for Student Aid** (FAFSA) on file with the UA Office of Scholarships and Financial Aid.

For more information about scholarships for continuing undergraduate students in the College of Engineering, please see [Department Scholarship Contacts](#).

Free Tuition for Arizona Native Americans

With the Arizona Native Scholars Grant, undergraduates from the state's 22 federally recognized Native American tribes do not pay tuition or mandatory fees at the University of Arizona Tucson campus. More details about the grant, how to apply, and other programs that serve Indigenous students are available on the [Scholarships and Financial Aid website](#).

Scholarships for Incoming Transfer Students

Transfer students make up approximately 20% of the incoming class each year. The College of Engineering recognizes that transfer students face unique challenges when choosing to move on to the next part of their educational journey. As such, the college has developed a suite of transfer-specific scholarships. Learn more on our [Transfer Admissions page](#).

Declaring Your Major

Incoming first-year students will be enrolled as Engineering, No Major Selected, until they are ready to **declare their major** and have met the requirements. The College of Engineering provides an exploratory course – Engineering 102 – to assist students in selecting the right degree program. The major declaration process is non-competitive, with no caps or quotas.

If you are also admitted to the University of Arizona **Honors College**, you may be given an opportunity prior to orientation to pre-select one of the College's 18 **Engineering Majors**.

Reapplying or Changing to the College of Engineering

If you are an incoming student accepted to a non-Engineering major and you wish to be re-evaluated for admission to the College of Engineering, please fill out the following **form**.

Not Accepted to the College of Engineering

Students who applied but were not admitted to the College of Engineering must submit new

qualifying information (for example: seventh- or eighth-semester high school transcripts, college/university transcripts, ALEKS math placement score) to the Office of Admissions before their requests will be considered.

Engineering Not Chosen on Application

Students who did not select Engineering, or for whom it was not a first choice, are not required to resubmit qualifying information. The college will review your initial UA application.

Currently Enrolled

If you are already enrolled at the UA, please see **Declaring an Engineering Major**.

What #ENGRCATS Need to Know About Math Placement

Engineers use math (and science) as the foundation of everything they do – and #ENGRCats are no different. Before you even begin your coursework on campus, you can take steps to prepare for a successful math journey.

- **If you have one, submit your ACT/SAT to UA.** Even though standardized testing *is optional*, you can use an ACT or SAT score to help determine your appropriate initial math placement. (**See the score chart here.**) **With or without a test score, you'll still be encouraged to take UA's math placement assessment.**
- **Learn about the ALEKS PPL.** The ALEKS is UA's math placement exam. Read all about the exam and what to expect **here**. Although our 4-year degree plans begin with Calculus I, it is far more important to begin your academic career in the correct math class; you may need, or want, to start in another course to increase strength and confidence in math. *We recommend taking the ALEKS PPL as soon it appears in your Next Steps Center.* This allows time to review and re-test, if desired.
- **Think ahead for AP or Dual Enrollment.** If you're currently enrolled in AP or dual enrollment coursework senior year, you may also utilize these scores and/or classes to gain credit toward your degree (and to help with determining your math placement, too). To see the full list of AP exam scores and affiliated UA courses, **visit this site**. You'll have the opportunity to submit these scores prior to your New Student Orientation. (And, you should plan to tell your advisor about

any expected AP or dual enrollment credit during your 1:1 virtual advising appointment at New Student Orientation, too.)

- **Brush up on some resources.** The goal of the ALEKS exam is to make sure you're being matched into the right course for you based on your most current preparation in math. This does not mean, however, that you can't practice or prepare before taking the exam! You can review **the study guide** on the Math website prior to the launch of the exam, or **check out some resources** put together by the College of Engineering here.

Admissions

All students apply to Engineering graduate programs through the **UA Graduate College**.

Please visit the following UA Web pages for details:

- **Admissions Requirements**
- **Application Process**
- **Costs and Fees**
- **Dates and Deadlines**
- **Request Information**

[Apply Now](#)

Financial Support

A number of **fellowships, scholarships**, grants, graduate assistantships and jobs are available to help you attain your graduate education goals in the UA College of Engineering. Graduate assistant positions are available through **College departments**. We encourage you contact the departments in which you are interested.

Visit the UA Graduate College for additional information on **funding resources** and **financial aid**.

All graduate students must file the Free Application for **Federal Student Aid** to be eligible for need-based grants, loans, scholarships, and work-study programs.

Fellowships and Scholarships

Fellowship and scholarship announcements of interest to UA College of Engineering graduate students are listed below.

NASA MUREP Proposals

The NASA Office of Education Minority University Research and Education Project, or MUREP, seeks independently conceived fellowship proposals from students pursuing graduate degrees in NASA-relevant STEM disciplines. Selected students perform research at their campuses during the academic year and at a designated NASA field center during an annual 10- to 15-week Center Based Research Experience, or CBRE. Funding is provided for a maximum of three years.

Research Associateship Programs

The National Academies of Sciences, Engineering and Medicine administers numerous NRC postdoctoral fellowships. Stipends for recent PhD recipients range from approximately \$42,000 to \$80,000.

Sandia National Labs Master's Fellowships

These fellowships fully fund master's of science degrees for exceptional bachelor's-level minority candidates, who join multidisciplinary R&D teams as regular full-time Sandia employees.

NSF Graduate Research Fellowships

The NSF Graduate Research Fellowship Program, or GRFP, helps ensure the vitality and diversity of the scientific and engineering workforce in the United States. The program provides awardees \$34,000 annually for three years. During tenured years, NSF Fellows at the UA receive full tuition, paid mandatory fees, individual health insurance, and a yearly \$500 travel grant.

BIO5 Internship Connections

The BIO5 Institute regularly posts internships and jobs geared to the Arizona bioindustry –for graduate and undergraduate students in the life sciences, engineering, business and computer and information sciences.

UA Institute of the Environment Travel Awards

These semiannual awards provide funding, up to \$500, for graduate students to travel to meetings, workshops, conferences or symposia and present their research or to do fieldwork.

Research Assistant and Teaching Assistant Positions

RA and TA positions are available through **College departments**. We encourage you to contact the departments in which you are interested.

Additional Funding Sources

Visit the UA Graduate College for additional information on **funding resources** and **financial aid**.

All graduate students must file the **Free Application for Federal Student Aid** to be eligible for need-based grants, loans, scholarships, and work-study programs.

College of Engineering

Program Details

The University of Arizona's College of Engineering has launched the Software Engineering PhD degree in response to the high demand for individuals trained in the software engineering discipline.

The degree program is critical in driving student success in a rapidly changing world and tackling essential problems at the edges of human endeavor.

As a student graduating with a PhD in Software Engineering, you will be better positioned to develop the skills and mindsets to be leaders in software development, computing, machine learning, ever-increasing automation and connectivity, human and intelligent systems, data science, and network sciences.

Through the PhD program, you'll demonstrate the ability to design, develop, test, integrate, and evaluate software applications/products/systems in diverse computing and engineering domains. You'll also be able to critically analyze and review published research results and other literature related to your area of study. You'll also demonstrate your ability to articulate all aspects of the research in your software engineering specialization area, describe and defend the significance of your work, explain your research methodologies, and summarize your findings.

The global software engineering market alone will be worth \$64 billion by 2025, and it is a vital part of a larger industry. Some factors behind this growth include increased automation in multiple sectors, the demand for cloud-based solutions, the Internet of Things, and an increased number of devices that can be used in daily life for convenience. Thus, pursuing a Software Engineering PhD will give you a competitive edge in this fast-growing industry.

[View detailed program information](#)

No GRE is required for this graduate degree program.

A minor is required for this program and will be determined by the student and advisor.

Students who do not have a degree equivalent to the University of Arizona Bachelor of Science degree in a computing-related program may be admitted into the graduate program but may be required to complete additional graduate-level pre-requisite courses prior to enrolling in some graduate courses.

Proficiency in one or more programming languages OR one to two years of industry experience in a software-related position is required.

Residents of some U.S. Territories may not be eligible. Please see our **[Eligibility & State Authorization page](#) for more information.*

Program Type

Doctor of Philosophy

College

College of Engineering

Career

Graduate

Program Description

The Software Engineering PhD program has a firm engineering foundation that encompasses discovery-based education utilizing an experiential learning approach. As part of the curriculum, students will complete projects in areas that emphasize software engineering, communication, teamwork, critical thinking, and engineering professionalism. Which focuses on the application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software. In the PhD program, students will also conduct novel research in many diverse software engineering related areas. The program's flexibility allows students to design their course of study / research from a diverse pool of courses and research opportunities in software and computer engineering domains such as web and mobile applications, embedded systems, cybersecurity, machine learning, systems, and other interdisciplinary areas.

Program Requirements

Minimum Credit Units

Completion requirement

63

Core Coursework Requirements

Completion requirement

SFWE 513 (3) – Software Engineering Research Methods

SFWE 507 (3) – Foundations of Software Engineering

Complete 9 units of additional core coursework:

SFWE 502 (3) - Software DevSecOps

SFWE 503 (3) - Software Project Management

SFWE 504 (3) - Software Requirements Analysis and Test

SFWE 505 (3) - Software Architecture and Design

SFWE 506 (3) - Distributed Computing

Students that have a BS in Software Engineering and have demonstrated a given Core course's competencies must consult/work with their faculty advisor to substitute the course with one of the SFWE 5xx level courses listed below.

- Choose 16 units non-dissertation course work which must be approved by the faculty advisor and the Director of Graduate Studies:

- o Any SFWE core courses not used to meet the Core coursework requirement can also be used as technical electives

- o SFWE 501 (3) - SW Assurance

- o SFWE 508 (3) –Data Mining

- o SFWE 509 (3) – Cloud Computing Principles and Practices

- o SFWE 510 (3) – Cloud Native Software Engineering

- o SFWE 511 (3) – Software for Industrial Control Systems

- o SFWE 512 (3) –Robotics

- o SFWE 513 (3) – Software Engineering Research Methods

- o CSE 501 (3) – Operating Systems

- o ECE 503 (3) - Probability and Random Processes for Engineering Applications

- o ECE 509 (3) –Cybersecurity Concept, Theory, Practice

- o ECE 513 (3) –Web Development and the IoT

- o ECE 523 (3) –Engineering Applications of Machine Learning and Data Analytics

- o ECE 562 (3) –Computer Architecture and Design

- o ECE 576A (3) - Engineering of Computer Based Systems

- o ECE 576B (3) - Embedded System Design and Optimization

- o ECE 579 (3) –Principles of Artificial Intelligence

- o SIE 533 (3) –Fundamentals of Data Science for Engineers

- o SIE 558 (3) –Model Based Systems Engineering

- o SIE 577 (3) – Introduction to Biomedical Informatics

Colloquium and Dissertation Research (20 units)

- SFWE 695A - Colloquium (2)

- SFWE 920 - Dissertation Research (18)

Elective Coursework

Completion requirement

None

Additional Requirements

Completion requirement

Specific requirements will be included in the SFWE Graduate Handbook.

Doctoral Qualifying Exam (DQE)

The purpose of the Doctoral Qualifying Examination (DQE) is to assess students' fundamental knowledge in specific SFWE domains and whether students have an integrated understanding in those domains as well as their readiness of a student to undertake advanced graduate work.

Doctoral Comprehensive Exam

The purpose of the DQE is to determine whether the student has developed sufficient background and expertise for research in the field of their planned dissertation. Successful completion of the examination leads to formal admission to PhD candidacy.

Doctoral Final Written Dissertation

The candidate should develop a written document that demonstrates all aspects of their research including significance of the work, a detailed review of relevant literature, methodologies employed and/or developed, significant findings from the work, a critical discussion of the findings, limitations, and the impact, and potential for future research.

Note: The faculty advisor defines the specific format of the written dissertation. The Graduate College provides general formatting guidelines:

<https://grad.arizona.edu/gsas/dissertationtheses/dissertation-and-thesisformatting-guides>

Final Oral Defense

When the doctoral candidate has met the rigor and standards of scholarship and has documented the research in a dissertation, the candidate will publicly defend the dissertation and answer any general questions related to their work. The exact time and place of the oral defense must be announced publicly at least two weeks in advance of the oral defense. The oral defense is facilitated by a faculty committee appointed by the Dean of the Graduate College in consultation with the major department and chaired by the faculty advisor. The presentation portion of the oral defense is open to the public. Following the public presentation and discussion, the candidate will participate in a closed meeting with the committee for further evaluation

Minor Requirements for Doctoral Students in this Program

Completion requirement

All minor coursework (12 units) can come from a single discipline or split between two disciplines (6 units from each discipline).

Student Handbook

Completion requirement

SFEMS - Software Engineering

Download as PDF

[Electrical & Computer Engr](#)

Graduate Degree Seeking

MS - Master of Science

Program Type

Master of Science

College

College of Engineering

Career

Graduate

Program Description

Software engineering is key to how we engage with digital devices, and skills in programming and design position you to pioneer advancements in this dynamic field. The master's program in Software Engineering not only broadens your understanding of existing software architecture across a variety of technologies, but it also teaches you how to build and adapt software, leveraging your software developer degree to emerging technologies. Opt for a thesis or non-thesis track. The thesis path features classes such as Cloud Computing Principles and Practices; Principles of Artificial Intelligence; and Web Development and the Internet of Things, while the non-thesis route includes classes like Data Mining and Robotics.

SFEMS - Software Engineering

Download as PDF

[Electrical & Computer Engr](#)

Graduate Degree Seeking

MS - Master of Science

Completion requirement

30

Completion requirement

Program Curriculum:

The MS Software Engineering requires 30 credit hours to complete

Students may elect either the Thesis or Non-Thesis Option

- Thesis route: 24 units of SFWE graded coursework + 6 units of SFWE 910 (Thesis)
- Non-Thesis route: 30 units of SFWE graded coursework

Required Core Courses (12 total):

- SFWE 507 (3) - Foundations of Software Engineering (NEW)

Complete 3 courses of additional Core coursework (*select 3 from the following list of courses*):

- SFWE 502 (3) - Software DevSecOps
- SFWE 503 (3) - Software Project Management
- SFWE 504 (3) - Software Requirements Analysis and Test
- SFWE 505 (3) - Software Architecture and Design
- SFWE 506 (3) – Distributed Computing

Completion requirement

Complete one of the two subplan options below.

Completion requirement

The Master's Thesis Committee must abide by the [Graduate College's committee requirements](#). A minimum of two SFWE eligible faculty members must serve on the committee. The third member may be a Special Member.

Completion requirement

Please refer to the [Graduate Student Handbook](#) for students who are pursuing this program of study.

Completion requirement

Minimum Credit Units

30

Core Coursework Requirements

See program requirements above.

Elective Coursework

The remaining SFWE graduate elective credits will be selected with the approval of an SFWE Faculty Advisor.

Non-Thesis Option:

Complete 18 units from the **Technical Computing Electives** list below or in a closely related computing field (must be approved by graduate studies committee).

- Any SFWE core courses not used to meet the 12-units of *Core* coursework requirement can also be used as technical electives
 - SFWE 501 (3) - SW Assurance
 - SFWE 508 (3) –Data Mining
 - SFWE 509 (3) – Cloud Computing Principles and Practices
 - SFWE 510 (3) – Cloud Native Software Engineering

- SFWE 511 (3) – Software for Industrial Control Systems
- SFWE 512 (3) –Robotics
- SFWE 513 (3) – Software Engineering Research Methods
- CSE 501 (3) – Operating System Design
- ECE 503 (3) - Probability and Random Processes for Engineering Applications
- ECE 509 (3) –Cybersecurity Concept, Theory, Practice
- ECE 513 (3) –Web Development and the IoT
- ECE 523 (3) –Engineering Applications of Machine Learning and Data Analytics
- ECE 562 (3) –Computer Architecture and Design
- ECE 576A (3) - Engineering of Computer Based Systems
- ECE 576B (3) - Embedded System Design and Optimization
- ECE 579 (3) –Principles of Artificial Intelligence
- SIE 533 (3) –Fundamentals of Data Science for Engineers
- SIE 558 (3) –Model Based Systems Engineering
- SIE 577 (3) – Introduction to Biomedical Informatics
- *Other courses may be added at the discretion of the faculty advisor and GSC, or as additional new SFWE courses not listed in section III. **New Courses Needed** are developed.*

Additional Requirements

The Master's Thesis Committee must abide by the [Graduate College's committee requirements](#). A minimum of two SFWE eligible faculty members must serve on the committee. The third member may be a Special Member.

Student Handbook

Please refer to the [Graduate Student Handbook](#) for students who are pursuing this program of study.

Completion requirement

Minimum Credit Units

30

Core Coursework Requirements

See program requirements above.

Elective Coursework

The remaining SFWE graduate elective credits will be selected with the approval of an SFWE Faculty Advisor.

Thesis Option:

Complete 12 units from the **Technical Computing Electives** listed below or in a closely related computing field (must be approved by graduate studies committee).

- Any SFWE core courses not used to meet the 12-units of *Core* coursework requirement can also be used as technical electives
 - SFWE 501 (3) - SW Assurance
 - SFWE 508 (3) –Data Mining
 - SFWE 509 (3) – Cloud Computing Principles and Practices
 - SFWE 510 (3) – Cloud Native Software Engineering
 - SFWE 511 (3) – Software for Industrial Control Systems
 - SFWE 512 (3) –Robotics
 - SFWE 513 (3) – Software Engineering Research Methods
 - CSE 501 (3) – Operating System Design
 - ECE 503 (3) - Probability and Random Processes for Engineering Applications
 - ECE 509 (3) –Cybersecurity Concept, Theory, Practice
 - ECE 513 (3) –Web Development and the IoT
 - ECE 523 (3) –Engineering Applications of Machine Learning and Data Analytics
 - ECE 562 (3) –Computer Architecture and Design
 - ECE 576A (3) - Engineering of Computer Based Systems
 - ECE 576B (3) - Embedded System Design and Optimization
 - ECE 579 (3) –Principles of Artificial Intelligence
 - SIE 533 (3) –Fundamentals of Data Science for Engineers
 - SIE 558 (3) –Model Based Systems Engineering
 - SIE 577 (3) – Introduction to Biomedical Informatics
 - *Other courses may be added at the discretion of the faculty advisor and GSC, or as additional new SFWE courses not listed in section III. **New Courses Needed** are developed.*

Also complete 6 units of thesis (SFWE 910)

Additional Requirements

The Master's Thesis Committee must abide by the [Graduate College's committee requirements](#). A minimum of two SFWE eligible faculty members must serve on the committee. The third member may be a Special Member.

Student Handbook

Please refer to the [Graduate Student Handbook](#) for students who are pursuing this program of study.

Completion requirement

Minimum Credit Units

30

Core Coursework Requirements

Program Curriculum:

The MS Software Engineering requires 30 credit hours to complete

Students may elect either the Thesis or Non-Thesis Option

- Thesis route: 24 units of SFWE graded coursework + 6 units of SFWE 910 (Thesis)
- Non-Thesis route: 30 units of SFWE graded coursework

Required Core Courses (12 total):

- SFWE 507 (3) - Foundations of Software Engineering (NEW)

Complete 3 courses of additional Core coursework (*select 3 from the following list of courses*):

- SFWE 502 (3) - Software DevSecOps
- SFWE 503 (3) - Software Project Management
- SFWE 504 (3) - Software Requirements Analysis and Test
- SFWE 505 (3) - Software Architecture and Design
- SFWE 506 (3) – Distributed Computing

Elective Coursework

The remaining SFWE graduate elective credits will be selected with the approval of an SFWE Faculty Advisor.

Non-Thesis Option:

Complete 18 units from the **Technical Computing Electives** list below or in a closely related computing field (must be approved by graduate studies committee).

Any SFWE core courses not used to meet the 12-units of *Core* coursework requirement can also be used as technical electives.

- SFWE 501 (3) - SW Assurance
- SFWE 508 (3) –Data Mining
- SFWE 509 (3) – Cloud Computing Principles and Practices
- SFWE 510 (3) – Cloud Native Software Engineering
- SFWE 511 (3) – Software for Industrial Control Systems
- SFWE 512 (3) –Robotics
- SFWE 513 (3) – Software Engineering Research Methods
- CSE 501 (3) – Operating System Design
- ECE 503 (3) - Probability and Random Processes for Engineering Applications
- ECE 509 (3) –Cybersecurity Concept, Theory, Practice
- ECE 513 (3) –Web Development and the IoT
- ECE 523 (3) –Engineering Applications of Machine Learning and Data Analytics
- ECE 562 (3) –Computer Architecture and Design
- ECE 576A (3) - Engineering of Computer Based Systems
- ECE 576B (3) - Embedded System Design and Optimization
- ECE 579 (3) –Principles of Artificial Intelligence
- SIE 533 (3) –Fundamentals of Data Science for Engineers
- SIE 558 (3) –Model Based Systems Engineering
- SIE 577 (3) – Introduction to Biomedical Informatics
- *Other courses may be added at the discretion of the faculty advisor and GSC, or as additional new SFWE courses not listed in section III. **New Courses Needed** are developed.*

Thesis Option:

Complete 12 units from the **Technical Computing Electives** listed below or in a closely related computing field (must be approved by graduate studies committee).

Any SFWE core courses not used to meet the 12-units of *Core* coursework requirement can also be used as technical electives.

- SFWE 501 (3) - SW Assurance
- SFWE 508 (3) –Data Mining
- SFWE 509 (3) – Cloud Computing Principles and Practices
- SFWE 510 (3) – Cloud Native Software Engineering

- SFWE 511 (3) – Software for Industrial Control Systems
- SFWE 512 (3) –Robotics
- SFWE 513 (3) – Software Engineering Research Methods
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- ECE 509 (3) –Cybersecurity Concept, Theory, Practice
- ECE 513 (3) –Web Development and the IoT
- ECE 523 (3) –Engineering Applications of Machine Learning and Data Analytics
- ECE 562 (3) –Computer Architecture and Design
- ECE 576A (3) - Engineering of Computer Based Systems

B.S. in Software Engineering

Catalog Year 2025-26

Below is the advised sequence of courses for this degree program on Main Campus as of 1/16/2025.

Official degree requirements and course prerequisites are in the University General Catalog; prerequisites are subject to change.

Course Number and Title Units Prerequisites/Enrollment Requirements

1 st Semester

MATH 122A/B or MATH 125 Calculus I with Applications 3/5 Appropriate Math Placement

*CHEM 151 Chemical Thinking I or CHEM 161/163 4 Appropriate Math Placement

ENGL 101 or 107 or 109H First-Year Composition 3

ENGR102A/B Introduction to Engineering or ENGR 102 3

ENGR102A & 102B: Pre- or Co-requisite of

MATH 112 or higher; First-Year Status;

College of Engineering Major; must not have

taken ENGR102

UNIV 101 Intro to the General Education Experience 1

Semester Total 14/16

2 nd Semester

MATH 129 Calculus II 3 MATH 122B or 125 with C or higher

SFWE 101 Introduction to Software Engineering 3 MATH 112 or higher with C or higher

ECE 101 Programming 1 4 MATH 112, 120R, 122B or 125 or placement at

the level of MATH 120R

ENGL 102 or 108 First-Year Composition 3 ENGL 101 or ENGL 107

*CHEM 152 Chemical Thinking II or CHEM 162/164 or MSE 110

Solid State Chemistry or MCB 181R/L Intro Biology I 4

CHEM 152 and MSE 110: CHEM 151 or 141/143

or 161/163; MCB 181R/L: Appropriate Math

Placement level

Semester Total 17

3 rd Semester

MATH 243 Discrete Mathematics in Computer Science 3 MATH 122B, 125 or 129 (transfer credit

allowed)

*PHYS 141 Introductory Mechanics or PHYS 161H 4 MATH 122B, 125 or Appropriate Math Placement Level

ECE 201 Programming 2 3 ECE 101 with C or better

General Education: Exploring Perspectives (Artist) 3

General Education: Exploring Perspectives (Humanist) 3

Semester Total 16

4 th Semester

SIE 277 Object-Oriented Modeling and Design 3 ECE 101 or CSC 110

ECE 274A Digital Logic 4 ECE 101; Concurrent enrollment or completion of MATH 129

SFWE 201 Software Engineering Sophomore Colloquium 1 SFWE 101 or ECE 101

PHYS 241 Introductory Electricity and Magnetism or PHYS 261H 4 PHYS 241 or 261H: PHYS 141 or 140 or 161H;

MATH 129 or Appropriate Math Placement

MATH/STAT Elective - Consult with faculty advisor 3 Appropriate Math Placement

Semester Total 15

*Each of the following foundational science courses satisfies the requirements for General Education: Exploring Perspectives

(Natural Scientist): CHEM 151 or 152 or 161 or 162; or PHYS 141 or 161H.

Advanced Standing is required for 300- and 400-level engineering courses (see your academic advisor for details).

Course Number and Title Units Comments

5 th Semester

SIE 305 Introduction to Engineering Probability and Statistics 3

SFWE 301 Software Requirements Analysis and Test (Fall Only) 3

SFWE 303 Data Persistence 3

ECE 369A Fundamentals of Computer Organization (Fall Only) 4

Semester Total 13

6 th Semester

SFWE 302 or 405 Software Architecture and Design (Spring Only) 3

SFWE 401 Software Assurance and Security (Spring Only) 3

ECE 311 Engineering Ethics and Contemporary Issues 1

CSC 355 Discrete Structures and Basic Algorithms or

CSC 345 Analysis of Discrete Structure 3

Technical Elective 3 Consult with faculty advisor

†General Education: Building Connections 3

Semester Total 16

7 th Semester

SFWE 403 Software Project Management (Fall Only) 3

ENGR 498A Interdisciplinary Capstone 3 Senior Status

General Education: Exploring Perspectives (Social Scientist) 3

Technical Elective 3 Consult with faculty advisor

†General Education: Building Connections 3

Semester Total 15

8 th Semester

SFWE 402 DevOps (Spring Only) 3

Technical Elective 1 - 3 Consult faculty advisor for course approval

ENGR 498B Interdisciplinary Capstone 3 Senior Status

Technical Elective 3 Consult with faculty advisor

†General Education: Building Connections 3

UNIV 301 General Education Portfolio 1

Semester Total 14/16

†Students should work closely with their academic advisor to select General Education: Building Connections courses; some course

work in the major, such as some Technical Elective courses, may also fulfill General Education: Building Connections requirements.

MASTER LIST (All Campus es)									See "Tech Elective Tracks" Tab						
Title (Choose 4)	Course	Units	Prerequisite	Enrollment Restrictions	Main - Historically Offered	Online - Historically Offered	Yuma - Historically Offered	AMP?							
Introduction to Biomedical Informatics	SIE 477	3	EC 175	Adv. Standing	F	Not Offered	Not Offered	Yes					X		
Designing Cyber Resilient Systems	SIE 473	3	SI 471	Adv. Standing	On hold	Not Offered	Not Offered	Yes							
INSuRE	SIE 472	3	SI 471	Adv. Standing	SP	SP	SP	Yes	X			X			

Systems Cyber Security Engineering	SIE 471	3	Non e	Adv. Standing	F	Not Offered	F	Yes				X			
Model Based Systems Engineering	SIE 458	3	SE 45 4A	Adv. Standing	F	Not Offered	SP	Yes						X	
The Systems Engineering Process	SIE 454A	3	Non e	Adv. Standing	F	Not Offered	F/SP	Yes						X	
Optimization for Machine Learning	SIE 449	3	SE 270, SE 340	Adv. Standing	SP	Not Offered	SP	Yes	X						
Fund of Data Science for Engineers	SIE 433	3	SE 305 or equivalent	Adv. Standing	SP	Not Offered	F	Yes	X				X		
Simulation Modeling & Analysis	SIE 431	3	SE 305	Adv. Standing	F/SP	Not Offered	F/SP/SU	Yes	X	X					
Engineering Stats	SIE 430	3	SE 305	Adv. Standing	F	Not Offered	F	Yes	X						
Technical Sales & Marketing	SIE 415	3	Non e	Adv. Standing	F/SP	Not Offered	F/SP	Yes					X		

Embedded Computer Systems	SIE 370	4	Non	Adv. Standing	SP	Not Offered	Not Offered	No							X	
Directed Research	SFW E 492	3	Non	Department Consent	F/SP/SU	F/SP/SU	F/SP/SU	Yes								
Software for Industrial Control Systems	SFW E 411	3	EC E 275	Adv. Standing	F	F	F	No							X	
Cloud Native (SP25: SIE 496)	SFW E 410	3	SFW E 302	Adv. Standing	F	F	F	Yes			X					
Principles of Cloud Computing	SFW E 409	3	EC E 369A or CS C 252	Adv. Standing	SP	Coming soon	SP	Yes			X					
Computational Linguistics	LING 438		Non	N/A	F	Not Offered	Not Offered	No								
Artificial Intelligence	ISTA 450	3	MATH 243 or consent	N/A	F/SP	Not Offered	Not Offered	No	X							

			of in str uc to r													
Data Warehousing and Analytics in the Cloud	IST A 43 1	3	N on e	N/A	F/SP/SU	F	Not Offered	N o						X		
Algorithms for Games	IST A 42 5	3		N/A	Coming soon	Not Offered	Not Offered	N o		X						
Intro to Machine Learning	IST A 42 1	3	(E S O C 21 4 or IS TA 11 6 or D AT A 36 3) and (IS TA 13 0 or CS C	N/A	F/SP	Not Offered	Not Offered	N o	X					X		

			110)													
Introduction to Web Design & Development	ISTA 329	3	None	F/SP	F/SP	Not Offered	No									
Data Engineering	ISTA 322	3	N/A	F/SP	F/SP	Not Offered	No						X			
Data Mining and Discovery	ISTA 321	3	N/A	F/SP/SU	F/SP/SU	Not Offered	No						X			

			of in str uc to r													
Applied Data Visualization	IST A 32 0	3	IS TA 11 6, ES O C 21 4, or eq ui va le nt ; or co ns en t of in str uc to r	N/A	F	F	Not Offered	N o						X		
Advanced Game Development	GA M E 45 2	3		N/A	F/SP	Not Offered	Not Offered	N o								

Game Artificial Intelligence	GA ME 426	3	IS TA 350 or EC E 373 or CS C 210 or in str uc to r co ns en t.	N/A	SP	Not Offered	Not Offered	No	X						
Introduction to Game Development with Unity	GA ME 351	3	EC E 175 or IS TA 130 or CS C 110	N/A	F/SP/SU	F/SP/SU	Not Offered	No							
Disruptive Technologies	ES OC	3	N on e	N/A	F/SP/SU	F/SP/SU	SP	No							

	318															
Tech Ventures	ENTR 400 (387)	36	Recommended: ENTR 29	N/A	SP	Not Offered	Not Offered	No								
Artificial Intelligence	ECE 479	37	EC E 373 or SI E 27	Adv. Standing	SP	Not Offered	Not Offered	Yes	X					X		
Fund of Computer Networks	ECE 478	30	EC E 275 & EC E 31	Adv. Standing	F	Not Offered	Not Offered	Yes				X				X
Computer Aided Logic	ECE 474A	35	EC E 274A & EC E 27	Adv. Standing	F	Not Offered	Not Offered	Yes							X	

Fund Info Theory & Ntwk Security	EC E 47 1	3	N on e	Adv. Standing	SP	Not Offered	Not Offered	Ye s				X			
Knowledge System Engineering	EC E 46 6	3	N on e	Adv. Standing	SP	Not Offered	Not Offered	Ye s							
Computer Architecture & Design	EC E 46 2	3	EC E 36 9A	Adv. Standing	SP	Not Offered	Not Offered	Ye s						X	
Digital Signal Processing	EC E 42 9	3	M aj or s; EC E 34 0A	Adv. Standing	F/SP	F	Not Offered	Ye s		X				X	
Intro to Deep Learning: An Engineering Perspective	EC E 42 5 (TE M P ap pr ov ed)	3	M AT H 12 2A /B , EC E 20 1 & EC E 31 0*	Adv. Standing	F	Not Offered	Not Offered	Li ke ly							
Web Development & the IoT	EC E 41 3	3	EC E 27 5	Adv. Standing	F	Not Offered	Not Offered	Ye s							X

Microprocessor Organization	EC E 37 2A	4	EC E 27 4A & (E CE 20 7 or 22 0) & *E CE 27 5	Adv. Standing	F/SP	SP	Not Offered	N o							X	
Computational Techniques	EC E 33 0B	3	M aj or s; M AT H 22 3, M AT H 25 4, P H YS 14 3, P H YS 24 31	Adv. Standing	SP	Not Offered	Not Offered	N o		X						

			& EC E 17 5												
Applications of Engineering Math	EC E 31 0*	4	M AT H 25 4 & (E CE 20 7 or 22 0) & EC E 27 5	Adv. Standing	F/SP	F	Not Offered	N o							
Cyber Warfare	CY BV 48 0	3	CY B V 32 6 and CY B V 38 5 or co ns ent t of	N/A	Not Offered	F/SP	F/SP	N o					X		

			in str uc to r.													
Advanced Analytics for Security Operations	CY BV 47 4	3	N on e	N/A	Not Offered	F/SP	F/SP	N o				X				
Violent Python	CY BV 47 3	3	N on e	N/A	Not Offered	F/SP	F/SP	N o				X				
Malware Threats & Analysis	CY BV 45 4	3	N on e	N/A	Not Offered	F/SP	F/SP	N o				X				
Cyber Threat Intelligence	CY BV 43 5	3	N on e	N/A	See Advisor	F/SP/SU	F/SP/SU	N o				X				
Active Cyber Defense	CY BV 40 0	3	N on e	N/A	See Advisor	F/SP	F/SP	N o				X				
Introduction to Cyber Operations	CY BV 38 5	3	N on e	N/A	See Advisor	F/SP/SU	F/SP/SU	N o				X				
Introduction to Security Scripting	CY BV 31 2	3	N on e	N/A	Not Offered	F/SP/SU	F/SP/SU	N o				X				
Windows Security Essentials	CY BV 30 3	3	N on e	N/A	Not Offered	F	F	N o				X				
Linux Security Essentials	CY BV 30 2	3	N on e	N/A	Not Offered	F	F	N o				X				

Artificial Intelligence	AP CV 471	3	CS C 35 or 345	N/A	Not Offered	F	F	No	X							
Comparative Programming Languages	CS C(V) 372	3	CS Major ; CS C 210	N/A	F/SP	F	F	No								
Intro to Computer Vision	CS C 477	3	CS Major ; CS C 252, CS C 335, CS C 345, CS C 352 & (M AT	CS Majors	SP	Not Offered	Not Offered	No	X							

			H 21 5 or M AT H 31 3)													
Database Design	CS C 46 0	3	CS M aj or ; CS C 33 5 & CS C 34 5	CS Majors	F/SP	SP/SU	SP	N o								
Compilers and Systems Software	CS C 45 3	3	CS M aj or ; CS C 25 2, 34 5, 35 2. CS C 37 2	CS Majors	F/SP	Not Offered	Not Offered	N o								

		co m m en de d														
Prin Operating Systems	CS C 45 2	3	CS M aj or ; CS C 25 2, CS C 34 5 & CS C 35 2	CS Majors	SP	SP	SP	No							X	
Data Visualization	CS C 44 4	3	CS M aj or ; CS C 33 5 & CS C 34 5	CS Majors	F/SP	Not Offered	Not Offered	No								

Software Engineering	CS C 43 6	3	CS Major ; CS C 25 2, CS C 33 5, CS C 34 5, and CS C 35 2, and	CS Majors	F	Not Offered	Not Offered	No							
Computer Graphics	CS C 43 3	3	CS Major ; CS C 25 2, 33 5, 34 5, 35 2, and	CS Majors	F/SP	Not Offered	Not Offered	No							

			M A T H 21 5 or 31 3)												
Comparitive Programming Languages	C S C 37 2	3	C S M a j o r ; C S C 21 0	C S Majors	S P	Not Offered	Not Offered	N o							
Systems Programming and Unix	C S C 35 2	3	C S M a j o r ; C S C 25 2, 33 5, 34 5, 35 2, an d (M A T H 21 5 or	C S Major; CSC 210 and CSC 252	F/SP	Not Offered	Not Offered	N o							

			31 3)													
Cloud Computing	CS C 34 6		CS C 12 0	CS Majors	F	Not Offered	Not Offered	N o				X				
Web Programming			EC E 17 5 or IS TA 13 0 or CS C 11	N/A	F/SP	Not Offered	Not Offered	N o								X

Get Set for Top Technology Jobs

Software engineers build the technological infrastructure for our increasingly connected world. They are in high demand to solve complex engineering problems across industries and government, working in areas such as:

- aerospace and space exploration
- automation and connectivity
- biomedical modeling and devices
- data, computing and networking
- human and intelligent systems
- natural and fabricated environments

With an interdisciplinary curriculum that covers large-scale product development and incorporates intensive real-world design, the University of Arizona BS in software engineering prepares students for some of the most coveted technology jobs in the world.

The 2021 U.S. News & World Report ranked software developer, or engineer, as the best technology job in the United States, and the second-best job across all categories.

Academic Focus Areas

Covering the Entire Software Development Lifecycle

Bolstered by intensive design experiences that reflect industry demand and focused on large-scale product and system development to meet societal needs, the University of Arizona BS in software engineering goes far beyond programming fundamentals and theoretical applications. The degree program – administered by the department of Systems & Industrial Engineering – is offered on the main campus in Tucson, [Arizona Online](#), and [Yuma Campus](#). Courses cover the entire software development lifecycle, or SDLC. Among them are:

- Software requirements analysis and testing
- Software architecture and design
- Software assurance
- Project management
- Cost estimation
- Digital logic
- Computer organization

Students choose electives – such as web and mobile applications, or embedded systems – from across many of the college's eight departments.

Importantly, students learn about software DevOps, Agile methodology, and continuous integration and deployment. They use open-source integrated development environments, operating systems and cloud-based infrastructures to gain the specialized software skills in highest demand by top employers.

Research Opportunities

Globally recognized faculty and students collaborate across the campus and around the world to tackle challenges in health care, cybersecurity, big data and transportation. Ongoing research involving software engineering includes:

- Autonomous systems and robotics
- Communications, coding and information theory
- Computer architecture and cloud/distributed computing
- Data analytics, informatics and machine learning

- Embedded systems
- Wireless networking, security and systems



Outside the Classroom

Real-World Learning

Software engineering students do design projects every semester. They get experience participating in clubs and design-build competitions as well as internships with longtime industry partners.

Additionally, valuable contacts are made in student chapters of professional organizations such as IEEE and IISE/INCOSE and at specially scheduled events with industry leaders.

Career Paths

Endless Possibilities

Software engineering graduates work in corporate and government settings as well as in research and as independent consultants. They become software and application architects and developers, project managers, network engineers, and business owners. Software engineers are in high demand across industries with employers such as Google, Amazon, Microsoft, Snap Inc., Medtronic, Johnson & Johnson, Raytheon, General Dynamics, Honeywell and Wells Fargo. The list, at home and abroad, is limitless.



Create a Career in the Tech Revolution

Find your future with three top-ranked University of Arizona Engineering programs – Electrical & Computer Engineering, Computer Science & Engineering, and Software Engineering – housed within a single department.

Three Flexible Degree Programs to Fit Your High-Tech Future



Computer Science & Engineering

Drive tomorrow's technology by mastering complex computer and software systems for any industry. Build everything from everyday smart devices to AI-powered platforms.

[Computer Science & Engineering](#)



Electrical & Computer Engineering

Get hands-on with smart systems, robotics, energy, AI, cybersecurity, IoT and quantum tech. Gain practical experience from day one, preparing for an impactful tech job.

[Electrical & Computer Engineering](#)



Software Engineering

Design, build and launch software for all fields, including aerospace, gaming and health care. Integrate design thinking, project management and engineering principles.

[Software Engineering](#)

Research That Makes a Difference

Create the technologies shaping society. From smart grids and autonomous vehicles to artificial intelligence, machine learning and cybersecurity, gain the skills to transform industries and improve lives. Major interdisciplinary research centers include:

- Arizona Research Institute for Solar Energy
- Broadband Wireless Access Center
- Center for Quantum Networks
- Center to Stream Healthcare in Place
- Cloud and Autonomic Computing Center
- Wireless Innovation towards Secure, Pervasive, Efficient, and Resilient Next G Networks



Build a Better World With Engineering

- AI and machine learning
- Biomedical/healthcare tech
- Computer systems and networks
- Cybersecurity
- Microelectronics
- Optics and photonics
- Quantum computing
- Robotics and autonomous tech
- Software and embedded systems
- Wireless and mobile systems

Real-World Learning for Top Jobs

The department's three programs boast a deep relationship with industry. From tech giants to agile startups, make connections for internships, research projects and your future career. Each program's curriculum is laser-focused on the tools you need to land your ideal job.

Undergraduate Programs

[Undergraduate Programs](#)

Graduate Programs

[Graduate Programs](#)

Meet the Faculty

Associate Professor Tosiron Adegbija

high-performance embedded computing • low-power embedded systems design

Professor Ali Akoglu

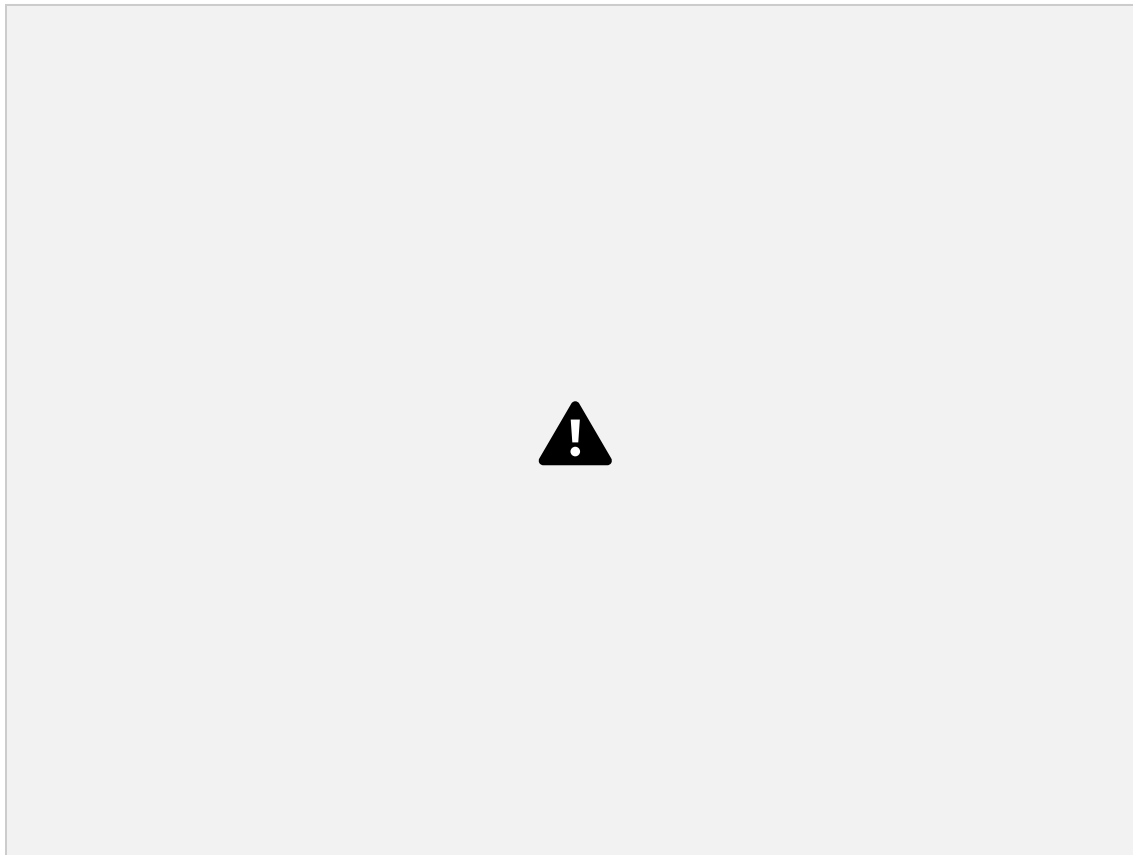
high-performance computing • reconfigurable computing • adaptive hardware systems

Associate Professor Boulat Bash

applying information theory to practical problems of reliability and security

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By the Numbers

Top 20

U.S. public research institution

(National Science Foundation and Times Higher Education)

#1

highest starting salary, computer engineering major

(U.S. News & World Report)

\$9 million

annual research expenditures (2025)

Top 10

best college majors for the future
(*Best College Reviews*)

Top 25

best computer engineering and electrical engineering graduate programs, public schools
ranking
(*U.S. News & World Report*)

4.5 out of 5

best graduate school programs
(*Money Magazine*)

Undergraduate Programs

Top Pick for Students With a Tech Mindset

Earn your bachelor's degree on campus or online in one of the University of Arizona's well-integrated, innovation-driven programs: [Computer Science & Engineering](#) (CSE), [Electrical & Computer Engineering](#) (ECE) or [Software Engineering](#) (SFWE).



Computer Science & Engineering

Drive tomorrow's technology by mastering complex computer and software systems for any industry. Build everything from everyday smart devices to AI-powered platforms.

[Computer Science & Engineering](#)



Electrical & Computer Engineering

Get hands-on with smart systems, robotics, energy, AI, cybersecurity, IoT and quantum tech. Gain practical experience from day one, preparing for an impactful tech job.

[Electrical & Computer Engineering](#)



Software Engineering

Design, build and launch software for all fields, including aerospace, gaming and health care. Integrate design thinking, project management and engineering principles.

[Software Engineering](#)

Academics That Reflect a Changing World

As technology continues to evolve at a rapid pace, the boundaries between computer, electrical and software engineering are becoming increasingly blurred. In today's interconnected world, complex systems demand seamless integration between hardware and software. That's why undergraduates in the ECE department pursue degrees that emphasize real-world application, technical depth and cross-disciplinary collaboration.

All engineering students at the U of A gain a solid foundation in analytical thinking, problem solving and technology plus essential skills in ethical leadership, communication and teamwork.

[Core classes](#) are interactive and incorporate discussion, one-on-one help, hands-on activities and real-life team projects.

Outside the Classroom

An invaluable part of the ECE student experience takes place in [student clubs](#) such as IEEE and Wildcat Robotics on interdisciplinary research and [design teams](#), and through participation in [internships](#). The success of ECE students in these activities is a big part of what makes them work ready and highly employable.

Job-Ready Graduates

Nearly every industry in every sector has a place for engineers from our department programs (Computer Science & Engineering, Electrical & Computer Engineering and Software Engineering), and constant advancements in technology keep them in demand. U of A ECE graduates are working at companies such as Google, Amazon, Intel, IBM, Qualcomm, Motorola, Microsoft, GM, Agilent Technologies, NASA, Boeing, Pixar and Texas Instruments, as well as Arizona-based companies such as Raytheon, Wells Fargo, Northrop Grumman and Honeywell.

Rankings

#1

highest starting salary, computer engineering major
(*U.S. News & World Report*)

Top 10

"best buy" public universities
(*Fiske Guide to College*)

#4

highest starting salary, electrical engineering major
(*U.S. News & World Report*)

Student in the Spotlight

Computation Crash Course

ECE undergrad [Celyn Jacobs](#) participated in a UA Research Experiences for Undergraduates summer program to conduct high-frequency long-range wireless communications.

"I can happily say that, after coming in with absolutely no knowledge of communications or artificial intelligence, I now have a firm understanding of how to develop, describe and implement complex systems involving neural networks, reinforcement learning and equalization."

Graduate Programs

Top Research University for Tech Engineering Degrees

University of Arizona graduate students in [Computer Science and Engineering](#) (CSE), [Electrical and Computer Engineering](#) (ECE), and [Software Engineering](#) (SFWE) are forward-thinking problem solvers who ultimately find themselves in high demand in academia, industry and government.



Computer Science & Engineering

Drive tomorrow's technology by mastering complex computer and software systems for any industry. Build everything from everyday smart devices to AI-powered platforms.

[Computer Science & Engineering](#)



Electrical & Computer Engineering

Get hands-on with smart systems, robotics, energy, AI, cybersecurity, IoT and quantum tech. Gain practical experience from day one, preparing for an impactful tech job.

[Flexible Degree Options with U of A Engineering Online](#)



Software Engineering

Design, build and launch software for all fields, including aerospace, gaming and health care. Integrate design thinking, project management and engineering principles.

[Get a Glimpse of Life in Tucson](#)

Here, you do not participate in research—you drive it. Whether you're advancing communications and signal processing or building next-generation computer systems and networks, you'll be part of groundbreaking work that makes a difference.

Choose a flexible, interdisciplinary curriculum and tailor your [MS](#) or [PhD](#) to your interests. [Online degree](#) options are also available for working engineers.

Questions? For more information on the U of A CSE, ECE and SFWE graduate programs, contact gradadmissions@ece.arizona.edu.

Highlights of the UA CSE, ECE and SFWE graduate programs include:

- Wide spectrum of interdisciplinary research
- Highly ranked programs
- Robust entrepreneurial community
- Strong industrial ties
- Tech and bioscience hotspot
- Globally recognized faculty
- Funding throughout degree lifecycle

High-Profile, Interdisciplinary Research

U of A Computer Science & Engineering, Electrical & Computer Engineering and Software Engineering graduate programs unite faculty and students from diverse disciplines and provide a number of high-profile research opportunities in the following [focus areas](#):

- Artificial intelligence and machine Learning
- Autonomous systems, communications and robotics
- Biomedical technologies
- Circuits, microelectronics and very-large scale integration
- Communications, coding and information theory
- Computer architecture and cloud/distributed computing
- Optics, photonics and terahertz devices and systems
- Quantum information science and engineering
- Signal, image and video processing
- Software engineering and embedded systems
- Wireless networking, security and systems

Student in the Spotlight

Artificial Brains for Real Inspiration

ECE doctoral student [Kama Svoboda](#) earned a U of A graduate fellowship and researches Spiking Neural Networks, an artificial intelligence that mimics the human brain.

"As a research scientist, my goal is to be a positive role model and mentor, especially for other women in STEM."

Research

Meeting the Challenge

From Better Healthcare to Safer Computing

Globally recognized ECE faculty are finding solutions to some of the world's biggest challenges – software for securing and managing big data in healthcare, imaging and blood tests to detect and diagnose disease, prostheses to save eyesight and robotic surgical tools, cyber DNA software to stop hackers in their tracks and devices that cloak wireless transmissions, for example.

The University of Arizona is one of the country's top public research institutions. Set in Tucson, a tech and bioscience hotspot, the U of A's Department of Electrical and Computer Engineering provides exceptional interdisciplinary research for students – students who say they don't just take part in research, they drive it.

With entrepreneurial ventures that boost Arizona's leadership in technology and top national and international companies and institutions for partners, ECE researchers are moving life-changing products out of the lab and into the marketplace with unprecedented speed.

Focus Areas

ECE – with annual external research expenditures exceeding \$7.6 million – houses multidisciplinary research centers and labs that support internationally recognized research in the following [primary areas](#):

- Artificial Intelligence and Machine Learning
- Autonomous Systems and Robotics Communications
- Biomedical Technologies
- Circuits, Microelectronics and Very-Large Scale Integration
- Communications, Coding and Information Theory
- Computer Architecture and Cloud/Distributed Computing
- Optics, Photonics and Terahertz Devices and Systems
- Quantum Information Science and Engineering

- Signal, Image and Video Processing
- Software Engineering and Embedded Systems
- Wireless Networking, Security and Systems

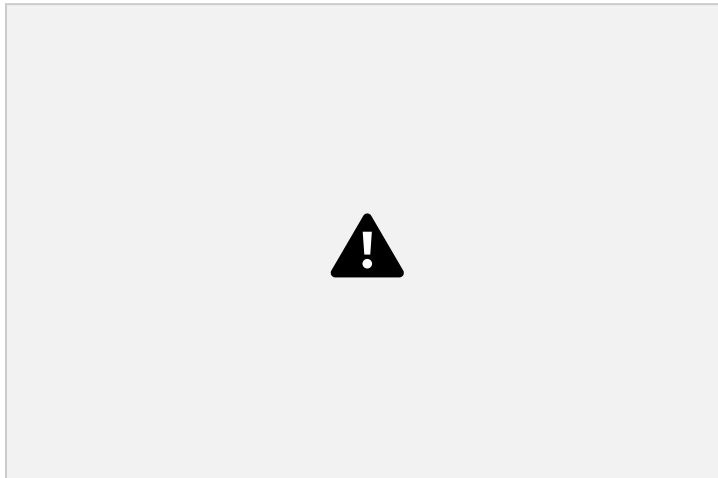
Project Highlight

A Positive Patient Process

ECE and neurosurgery professor [Allan Hamilton](#) is part of a team developing an artificial intelligence bot to train medical students to communicate with patients effectively and compassionately.

"This represents a step big forward in our medical education, utilizing the power of artificial intelligence to coach and evaluate medical students in a way that was previously unimaginable."

Image



Research Centers

U of A Electrical and Computer Engineering values collaboration across disciplines, units and institutions. ECE faculty direct the following collaborative [research centers](#):

- Arizona Research Institute for Solar Energy
- Broadband Wireless Access Center
- Center for Quantum Networks
- Center to Stream Healthcare In Place
- Cloud and Autonomic Computing Center
- Wireless Innovation towards Secure, Pervasive, Efficient, and Resilient Next G Networks

Featured Videos

Computer Science & Engineering

Engineer Tomorrow's Tech

Blend software and hardware to power the next generation of intelligent systems.

Computer Science & Engineering (CSE) at the U of A finds holistic solutions to improve the world. Whether you're creating AI systems from the ground up, designing embedded devices, or building secure cloud platforms. CSE at Arizona prepares you to lead in a rapidly evolving digital world.

AI-Driven, Systems-Oriented Curriculum

Go beyond coding. Students design full-stack computing systems, from microcontrollers to machine learning algorithms, integrating software and hardware to solve complex challenges. Degrees are offered for [BS](#), [MS](#), and [PhD](#), including an accelerated master's program.

Get a snapshot of [Computer Science & Engineering \(PDF\)](#)



Faculty Leading the Future

Study with internationally recognized experts in:

- Artificial intelligence and machine learning
- Cybersecurity
- Biomedical technology
- Quantum computing
- Robotics and autonomous systems
- Cloud architecture
- Embedded devices

Online Programs

Fully online [BS](#) and [MS](#) degrees are available, offering flexible learning options that fit your busy schedule.



Real-World Experience to Make a Difference

Collaborate on capstone design projects with companies like Raytheon, Honeywell and NASA to develop industry solutions. And students at all levels take part in game-changing research, including AI, wireless systems, biomedical tech and quantum computing.

Build leadership and community through student organizations, such as IEEE, Robotics Club, SWE, SHPE, NSBE, AISES and oSTEM.

Careers in High Demand

Graduates of the CSE program are equipped for success in fast-growing fields like:

- Artificial intelligence and data science
- Cybersecurity and cloud architecture
- Autonomous vehicles and smart infrastructure
- Biomedical and aerospace systems
- Software engineering and DevOps

Projected growth in AI-related jobs exceeds 20% by 2030. The median salary for CSE professionals is \$97,000.

Top employers include:

Intel • Microsoft • Google • Amazon • Medtronic • Northrop Grumman • IBM

Get a Head Start on the Future

Explore undergraduate and graduate programs in computer science and engineering at the University of Arizona to lead the next wave of computing.

Electrical & Computer Engineering

Power Up Your Future

Design and integrate the systems that drive today's most advanced technologies – from smart energy grids and autonomous vehicles to AI-powered healthcare and secure cloud computing.

Electrical & Computer Engineering (ECE) at the U of A tackles major challenges, such as the Internet of Things, big data, mobile health, cybersecurity, and wireless systems. Whether on campus or online, students gain the knowledge and skills to lead the next wave of technology.

Image



Student-Centered Academics

Students enjoy a collaborative learning environment that combines hands-on labs, team-based projects and personalized instruction, equipping them to work across industries where hardware, software, and intelligent systems intersect. Degrees are offered for [BS](#), [MS](#), and [PhD](#), including an accelerated master's program.

Specializations include:

- Autonomous systems and robotics
- Biomedical technologies
- Circuits, microelectronics and VLSI
- Communications, coding and information theory
- Computer architecture and cloud computing

- Optics, photonics and terahertz systems
- Signal, image and video processing
- Software engineering and embedded systems
- Wireless networking, security and systems

Online Programs

Fully online [BS](#) and [MS](#) degrees are available, offering flexible learning options that fit your busy schedule.

Research That Shapes the Future

With a legacy spanning more than a century, ECE has helped power every major wave of modern technological advances. Today, researchers are leading the charge in:

- **Real-time biomedical monitoring** via wireless and wearable technologies
- **Smart diagnostics** using sensors and big data to detect diseases earlier
- **Autonomous energy systems**, including next-generation smart grids
- **Cybersecurity solutions** that protect the nation's critical infrastructure
- **Quantum and optical systems**, for ultra-fast communication and sensing
- **AI modeling and edge intelligence** to supercharge smart environments

Students at every level do real research through labs, faculty-led projects and independent study.

Learning Beyond the Classroom

Hands-on experience is at the heart of the U of A ECE education. Students participate in national competitions, interdisciplinary design teams, and professional organizations like IEEE, SWE and the Arizona Solar Racing Team.

In the Interdisciplinary Capstone design course, seniors collaborate with industry sponsors to develop innovative, real-world solutions to engineering challenges.

High-Demand Career Opportunities

ECE graduates are in demand across industries – from aerospace and defense to healthcare, energy and consumer tech. Many students receive job offers before graduation, and alumni hold leadership roles around the world.

Top employers include:

Raytheon • Intel • IBM • Qualcomm • Microsoft • Honeywell • Texas Instruments • NASA • Boeing • Pixar • Medtronic • General Motors

Ready, Set, Power Up

Explore undergraduate and graduate programs in electrical and computer engineering at the University of Arizona.

Software Engineering

Build Global Solutions

Design, develop and maintain complex, high-impact software systems for any field.

Software Engineering (SFWE) at the U of A prepares students to build the programs that run our world. Take on every stage of software production – from theory and ideation to lifecycle development, DevOps, agile methodologies, continuous integration and deployment – to create a better digital world.

Curriculum Shaped by Industry

Since its launch in 2021, the program has been guided by top industry partners to meet modern demands. Courses focus on large-scale software product development and system-level design, giving graduates the tools to create solutions that scale. Classes are available on the main Tucson campus, [Yuma](#), [Chandler](#), or fully online. Degrees are offered for [BS](#), [MS](#), and [PhD](#), including an accelerated master's program.

Image



Hands-On Experience for Job Success

Graduates are equipped for success in virtually any area, including:

- Aerospace
- Artificial intelligence
- Health care technology
- Smart infrastructure
- Financial systems
- Data science
- Automation

Online Programs

Fully online [BS](#), [MS](#) and [PhD](#) degrees are available, offering flexible learning options that fit your busy schedule.

High-Demand Career Skills

Graduates are ready to jump into rewarding jobs with companies like Google, Raytheon, Intel, Microsoft, Northrop Grumman, Snap Inc. Honeywell and Amazon, along with national research centers such as Jet Propulsion Laboratory, Sandia and Oak Ridge.

These partners serve as program mentors, ensuring students have experience in the latest tools, cloud infrastructures, and open-source platforms.

Take the Next Step in Software Engineering

Launch your career in one of the most in-demand and future-focused fields.

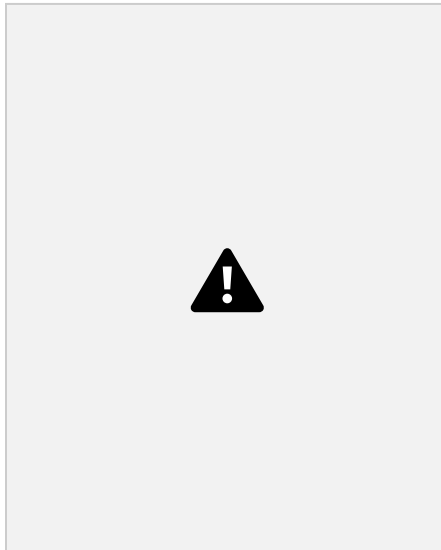
Undergraduate Programs

[Undergraduate Programs](#)

Graduate Programs

[Graduate Programs](#)

Image



"Software engineering is one of the fastest-growing fields, with careers expected to grow over 25% in the next decade. By 2025, the global market will top \$64B, driven by automation, AI/ML, cloud computing, and smart devices. For students, this means more opportunities than ever to build innovative, impactful careers in a wide range of industries."

Sharon O'Neal, Founder of the Software Engineering Program and Professor of Practice

By the Numbers

1

tech job in U.S.
(*U.S. News & World Report*)

\$ 110,000 +

median salary
(*Bureau of Labor Statistics*)

#5

most affordable online software engineering degree, public universities
(*OnlineU.com*)

Undergraduate Degrees

Innovative Tech Degrees for the Future

The ECE Department offers three Bachelor of Science degrees: [Computer Science & Engineering](#) (CSE), [Electrical & Computer Engineering](#) (ECE) and [Software Engineering](#) (SFWE) with both on-campus and online options.

BS in Computer Science & Engineering

Computer science and engineering majors create computing technologies that shape our interconnected world. They are in high demand across industries and government, focusing in areas such as:

- Artificial intelligence and machine learning
- Cybersecurity
- Biomedical technology
- Quantum computing
- Robotics and autonomous systems
- Cloud architecture
- Embedded devices

With an interdisciplinary curriculum and intensive real-world design, the University of Arizona [BS in CSE program](#) prepares students for some of the world's top technology jobs.

BS in Electrical & Computer Engineering

The ECE curriculum allows students to focus on the electronic foundations of our connected world, from microchips to communication networks. Highly sought after across industries, electrical and computer engineers drive breakthroughs in areas such as:

- Autonomous systems and robotics
- Biomedical technologies
- Circuits, microelectronics and VLSI
- Communications, coding and information theory
- Computer architecture and cloud computing
- Optics, photonics and THz devices and systems
- Signal, image and video processing
- Software engineering and embedded systems
- Wireless networking, security and systems

The University of Arizona [BS in ECE program](#) integrates circuit analysis, digital systems, and practical design projects, positioning graduates for leading roles in technology innovation.

BS in Software Engineering

Software engineers power the digital world building the systems that connect people, data, and technology. With high demand across every sector, they lead innovation in areas such as:

- Aerospace
- Artificial intelligence
- Health care technology
- Smart infrastructure
- Financial systems
- Data science
- Automation

The University of Arizona [BS in SFWE](#) program covers large-scale product development and intensive real-world design preparing students for some of the most in-demand technology jobs.

Online BS Option

The ECE Department has flexible, part-time online options for students pursuing a BS in [Computer Science & Engineering](#), [Electrical & Computer Engineering](#) or [Software Engineering](#).

Accelerated Master's Program

The U of A Graduate College gives exceptional ECE department students the option of earning a bachelor's degree and a master's degree in as few as five years. See the [CSE](#), [ECE](#) and [SFWE](#) Accelerated Master's Programs for more information.

Get a snapshot of the CSE, ECE and SFWE [Accelerated Master's Programs \(PDF\)](#).

Curriculum

Students interested in electrical and computer engineering often focus on electronic systems, signal processing, control and telecommunications. Computer science and engineering students typically concentrate on software design, computer chips and robotics. Software engineering students emphasize large-scale software development and system design. Together, these three areas are well integrated, providing a comprehensive foundation across hardware and software disciplines.

Four-Year Degree Plans

The College of Engineering's on-campus [four-year degree plans](#) provide the advised sequence of courses for a BS in computer science & engineering, electrical & computer engineering and software engineering.

Flow Charts

Major requirements, available as PDFs, help students plan around their particular interests.

Intro Mech Elect. & Magn. Object-Oriented

Modeling

Software Req.

Analysis

Software

Assurance

Software

Project Mng.

DevSecOps

PHYS 141 PHYS 241 SIE 277 SFWE 301* SFWE 401* SFWE 403* SFWE 402*

(4) (4) (3) (3) Fall Only (3) SPRING ONLY (3) Fall Only (3)

Intro to

Engineering

ENGR 102 or

ENGR 102A/B

(3)

Bachelor of Science, Software Engineering

(Revised 2025)

Catalogs 25-26

- Students must earn a grade of C or higher in all required engineering courses. These courses are noted with a “*” symbol.
- Students must complete four technical electives. Technical electives are upper division 300/400 level courses.
- Students must declare advanced standing status between their sophomore and junior semesters in order to take 300-level

engineering courses. Students must maintain a minimum of 2.5 GPA within their engineering coursework.

Freshman Sophomore Junior Senior

Calculus I Calculus II Discrete Math MATH Elect. Intro to Engr. Software Interdiscpl Interdiscpl

Math 122A/B or Math 223, 254, Prob. & Stats. Architecture Design Design

Math 125 Math 129 Math 243 or 313 SIE 305* SFWE 405* ENGR 498A ENGR 498B

(5) or (3) (3) (3) (3) (4) (3) Spring Only (3) (3)

Chemical Chemical Intro to Software Exploring Engr Ethics Building Building Building

Thinking I Thinking II Engineering: Perspective: Connection Connection Connection

CHEM 151 CHEM 152 SFWE 101* Humanist ECE 311*

(4) (4) (3) (3) (1) (3) (3) (3)

Intro to General

UNIV 101

(1)

On-Campus Graduate Degrees

Flexible Options to Match Your Goals

The U of A Department of Electrical & Computer Engineering offers PhD and MS degrees in Computer Science & Engineering, Electrical & Computer Engineering and Software Engineering including ECE and SFWE Accelerated Master's degrees. A customizable curriculum lets you focus on your interests!

[Online options](#) are also available.

Follow the links on this page to the U of A Graduate College for more information. Additionally, the ECE Graduate Program Handbook expands on program requirements and expectations.

Doctor of Philosophy Programs

- Direct track immediately following bachelor's degree
- Post-MS track

Computer Science & Engineering

The PhD Computer Science & Engineering curriculum applies computer science theory and software development fundamentals to produce computing-based solutions. It includes

substantial coverage of algorithms and complexity, computer science theory, concepts of multiple programming languages, software development, and engineering principles.

Electrical & Computer Engineering

PhD students in the Electrical & Computer Engineering program don't just participate in research - these visionary problem-solvers lead the charge, developing new technology and innovating better systems for wireless communication, smart systems, big data, cloud computing, solar energy and more. The program draws from coursework and hands-on, interdisciplinary research, and also connects the student to an entrepreneurial community. In addition, as a tech and bio hub, the University of Arizona maintains long-standing partnerships with industry titans such as Raytheon Technologies, Intel, IBM, Boeing, Pixar, NASA, Microsoft, Honeywell International, General Motors and Texas Instruments.

Software Engineering

The Software Engineering PhD program has a firm engineering foundation that encompasses discovery-based education utilizing an experiential learning approach. As part of the curriculum, students will complete projects in areas that emphasize software engineering, communication, teamwork, critical thinking and engineering professionalism.

Master of Science Programs

- Thesis
- Non-thesis
- Accelerated option available for current U of A undergraduates

Computer Science & Engineering

The MS Computer Science & Engineering curriculum applies computer science theory and software development fundamentals to produce computing-based solutions. It includes substantial coverage of engineering principles applied to the design of large, networked, scalable computer systems. Competencies include algorithms and complexity, concepts of multiple programming languages, software development, real time, embedded, and IoT systems design and other broad based engineering principles.

The program has a firm engineering foundation that encompasses discovery-based education utilizing an experiential learning approach. Students will complete projects in areas that emphasize computing theory, communication, teamwork, critical thinking, and engineering professionalism. The MS program's flexibility allows students to design their course of study/research from a diverse pool of courses and research opportunities in software, computer

science and computer engineering domains such as web and mobile applications, embedded systems, cybersecurity, machine learning, Quantum computing, systems, and other interdisciplinary areas.

Computer Science & Engineering Accelerated Master's Program

The MS Computer Science and Engineering curriculum applies computer science theory and software development fundamentals to produce computing-based solutions. It includes substantial coverage of engineering principles applied to the design of large, networked, scalable computer systems. Competencies include algorithms and complexity, concepts of multiple programming languages, software development, real time, embedded, and IoT systems design and other broad based engineering principles.

The program has a firm engineering foundation that encompasses discovery-based education utilizing an experiential learning approach. Students will complete projects in areas that emphasize computing theory, communication, teamwork, critical thinking, and engineering professionalism. The MS program's flexibility allows students to design their course of study/research from a diverse pool of courses and research opportunities in software, computer science and computer engineering domains such as web and mobile applications, embedded systems, cybersecurity, machine learning, Quantum computing, systems, and other interdisciplinary areas.

Electrical & Computer Engineering

The MS degree program in Electrical & Computer Engineering suits working engineers seeking to excel in the creation and planning of new technologies. A faculty of groundbreaking leaders in disciplines ranging from communications and signal processing to advanced computer systems collaborates with students on a flexible, customizable curriculum. Gain a versatile spectrum of knowledge from multiple technical fields or complete a focused academic plan centered on select subjects. Choose coursework in areas such as: remote sensing; high-performance computing; holography and diffractive optics; simulation modeling and heterogeneous systems design; wireless communications and networks; and detection and estimation.

Electrical & Computer Engineering Accelerated Master's Program

The Electrical & Computer Engineering master's program is designed for working engineers aiming to lead in developing new technologies. Work with expert faculty on a flexible, customizable curriculum—broad or focused—covering areas like remote sensing, high-performance computing, holography, simulation modeling, wireless networks, and detection and estimation.

Software Engineering

Software engineering is key to how to engage with digital devices, and skills in programming and design to pioneer advancements in this dynamic field. The MS program in Software Engineering broadens one's understanding of existing software architecture across a variety of technologies and teaches one how to build and adapt software, leveraging the software developer degree to emerging technologies.

Software Engineering Accelerated Master's Program

The Software Engineering master's program is designed for working professionals aiming to lead in creating reliable, scalable software systems. Work with expert faculty on a flexible, customizable curriculum covering areas like software architecture, cloud computing, AI and machine learning, cybersecurity, embedded systems and large-scale system integration.

Courses

See the Department of ECE's [graduate courses](#) and associated syllabi for detailed information.

ECE Courses

ECE 101: Programming I

Enrollment Requirements: Math 112, 120R, 122B, or 125, or placement at the level of Math 120R.

[Course details](#)

ECE 201: Programming II

Enrollment Requirements: ECE 101

[Course details](#)

ECE 207: Elements of Electrical Engineering

Enrollment Requirements: Prerequisite or concurrently enrolled in: MATH 122B or MATH 125.

[Course details](#)

ECE 220: Basic Circuits

Enrollment Requirements: MATH 129 and (PHYS 241 or PHYS 251 or PHYS 261H)

[Course details](#)

ECE 274A: Digital Logic

Enrollment Requirements: ECE 101. Prerequisite or concurrent enrollment in MATH 129.

[Course details](#)

ECE 304A: Design of Electronic Circuits

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 351C.

[Course details](#)

ECE 304C: Theory of Computation

Enrollment Requirements: Engineering. ECE 201.

[Course details](#)

ECE 310: Applications of Engineering Mathematics

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 101 and MATH 254.

[Course details](#)

ECE 311: Engineering Ethics and Contemporary Issues

Enrollment Requirements: Advanced Standing: Engineering. ECE majors only.

[Course details](#)

ECE 320A: Circuit Theory

Enrollment Requirements: Advanced Standing: Engineering. Major or minor: ECE. MATH 254 and ECE 220.

[Course details](#)

ECE 330B: Computational Techniques

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. MATH 223 and MATH 254 and PHYS 143 and PHYS 241 and ECE 101.

[Course details](#)

ECE 340A: Introduction to Communications

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 320A.

[Course details](#)

ECE 351C: Electronic Circuits

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 320A.

[Course details](#)

ECE 352: Device Electronics

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. Prerequisite or concurrent enrollment in ECE 351C.

[Course details](#)

ECE 369A: Fundamentals of Computer Organization

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 101 and ECE 274A.

[Course details](#)

ECE 372A: Microprocessor Organization

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 201 (ECE 207 or 220) and ECE 274A (concurrent enrollment in ECE 201 okay).

[Course details](#)

ECE 373: Object-Oriented Software Design

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 201.

[Course details](#)

ECE 381A: Introductory Electromagnetics

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. MATH 223 and ECE 220.

[Course details](#)

ECE 401C: Data Management

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE or CSE. Prerequisite ECE 201 and prerequisite or concurrent enrollment in CSC 355.

[Course details](#)

ECE 402C: Operating System Design

Enrollment Requirements: Advanced Standing: Engineering. Majors: CSE or ECE. ECE 201 with a grade of C or better.

[Course details](#)

ECE 403A: Mathematical Methods for Optics & Photonics

Enrollment Requirements: Major: OSE. Advanced Standing: Engineering. MATH 322. Concurrent with OPTI 403A.

[Course details](#)

ECE 403C: Algorithms Design and Analysis for Engineering Applications

Enrollment Requirements: Advanced Standing: Engineering. ECE 201 and CSC 355.

[Course details](#)

ECE 404: Optical Spectroscopy of Materials

Enrollment Requirements: Advanced Standing: Engineering. Concurrent with MSE 404.

[Course details](#)

ECE 405C: Advanced Data Structures

Enrollment Requirements: Advanced Standing: Engineering. ECE 201 and ECE 401C.

[Course details](#)

ECE 406C: Compiler Design

Enrollment Requirements: Advanced Standing: Engineering. ECE 201 and ECE 304C.

[Course details](#)

ECE 407: Digital VLSI Systems Design

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 274A and ECE 351C.

[Course details](#)

ECE 408: Agent-Based Simulation

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 175.

[Course details](#)

ECE 411: Numeric Modeling of Physics & Biological Systems

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE.

[Course details](#)

ECE 413: Web Development and the Internet of Things

Enrollment Requirements: Advanced Standing: Engineering. ECE 201.

[Course details](#)

ECE 414A: Photovoltaic Solar Energy Systems

Enrollment Requirements: Advanced Standing: Engineering.

[Course details](#)

ECE 417: Measurement and Data Analysis in Biomedical Engineering

Enrollment Requirements: Senior status only. Concurrent with BME 417.

[Course details](#)

ECE 425: Introduction to Deep Learning: An Engineering Perspective

Enrollment Requirements: Advanced Standing: Engineering. ECE 201, ECE 310, MATH 125 or MATH 122B.

[Course details](#)

ECE 429: Digital Signal Processing

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 340A.

[Course details](#)

ECE 430: Optical Communication Systems

Enrollment Requirements: Advanced Standing: Engineering. Concurrent with OPTI 430.

[Course details](#)

ECE 434: Electrical and Optical Properties of Materials

Enrollment Requirements: Advanced Standing: Engineering.

[Course details](#)

ECE 435A: Digital Communications Systems

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 340A.

[Course details](#)

ECE 441A: Automatic Control

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 320A.

[Course details](#)

ECE 442: Digital Control Systems

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 320A.

[Course details](#)

ECE 450: Analog Integrated Circuits

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 351C.

[Course details](#)

ECE 451A: Introduction to Physical Electronics

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 381A.

[Course details](#)

ECE 455: Introduction to Quantum Mechanics and Quantum Information Processing

Enrollment Requirements: Advanced Standing in Engineering. MATH 254 and ECE 275 and (ECE 207 or ECE 220) and ECE 310.

[Course details](#)

ECE 456: Optoelectronics

Enrollment Requirements: ECE 381A

[Course details](#)

ECE 459: Fundamentals Of Optics For Electrical Engineers

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 381A.

[Course details](#)

ECE 462: Computer Architecture and Design

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 369A.

[Course details](#)

ECE 463C: Introduction to Extended Reality (XR)

Enrollment Requirements: Advanced Standing: Engineering. ECE 201 and ECE 310.

[Course details](#)

ECE 466: Knowledge-System Engineering

Enrollment Requirements: Major: ECE. Advanced Standing: Engineering.

[Course details](#)

ECE 467: Computer Graphics and CAD

Enrollment Requirements: Major: ECE. Advanced Standing: Engineering.

[Course details](#)

ECE 471: Fundamentals of Information and Network Security

Enrollment Requirements: ECE 201 and ECE 310.

[Course details](#)

ECE 472: Design, Modeling, and Simulation for High Technology Systems in Medicine

Enrollment Requirements: Recommendations and additional information: Mastery of material in the areas of electrical and computer engineering, or mechanical engineering, or biomedical engineering at the level commensurate with senior/junior level standing.

[Course details](#)

ECE 473: Software Engineering Concepts

Enrollment Requirements: Major: ECE. ECE 373.

[Course details](#)

ECE 474A: Computer-Aided Logic Design

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 201 and ECE 274A.

[Course details](#)

ECE 478: Fundamentals of Computer Networks

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 201 and ECE 310.

[Course details](#)

ECE 479: Principles of Artificial Intelligence

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 373.

[Course details](#)

ECE 484: Antenna Theory and Design

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 381A

[Course details](#)

ECE 486: Microwave Engineering I: Passive Circuit Design

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 381A.

[Course details](#)

ECE 488: Microwave Engineering II: Active Circuit Design

Enrollment Requirements: Advanced Standing: Engineering. Major: ECE. ECE 486.

[Course details](#)

SFWE Courses

SFWE 101: Introduction to Software Engineering

Enrollment Requirements: MATH 122B or MATH 125

[Course detail](#)

SFWE 201: Software Engineering Sophomore Colloquium

Enrollment Requirements: SFWE 101

[Course details](#)

SFWE 301: Software Requirements Analysis and Test

Enrollment Requirements: Advanced Standing and ECE 201

[Course details](#)

SFWE 303: Data Persistence

Enrollment Requirements: Advanced Standing: Engineering. ECE 275 or ECE 201.

[Course details](#)

SFWE 401: Software Assurance & Security

Enrollment Requirements: ECE 201

[Course details](#)

SFWE 402: DevSecOps

Enrollment Requirements: ECE 201 recommended prior to enrollment.

[Course details](#)

SFWE 403: Software Project Management

Enrollment Requirements: SFWE 301

[Course details](#)

SFWE 405 - Software Architecture and Design

Enrollment Requirements: Advanced Standing: SFWE 301 or CSC 355 or CSC 345.

[Course details](#)

SFWE 407: Foundations of Software Engineering

Enrollment Requirements: SFWE 301

[Course details](#)

SFWE 409 Principles of Cloud Computing

Enrollment Requirements: Advanced Standing: Engineering and (ECE 369A or CSC 252).

[Course details](#)

SFWE 410: Cloud-Native

Enrollment Requirements: Advanced Standing: Engineering. SFWE 406 or SFWE 302.

[Course details](#)

SFWE 411: Software for Industrial Control Systems

Enrollment Requirements: ECE 201, SFWE 401

[Course details](#)

SFWE 491: Software Engineering Preceptor

Enrollment Requirements: Department consent is required to enroll in this course.

[Course details](#)

1. [Home](#)
2. [Research](#)
3. Focus Areas

Focus Areas

Artificial Intelligence and Machine Learning

- [Ehsan Azimi](#)
- [Boulat Bash](#)
- [Ali Bilgin](#)
- [Siyang Cao](#)
- [Jingdi Chen](#)
- [Jyotikrishna Dass](#)
- [David Ebert](#)
- [Marwan Krunz](#)
- [Ming "Fred" Li](#)
- [Bo Liu](#)
- [Xiaolong Ma](#)
- [Soheil Salehi](#)
- [Pratik Satam](#)
- [Ravi Tandon](#)
- [Changlong Wu](#)
- [Michael Wu](#)
- [Han Xu](#)
- [Huanrui Yang](#)



Autonomous Systems, Communications and Robotics

- [Ehsan Azimi](#)
- [Siyang Cao](#)
- [Ming "Fred" Li](#)
- [Bo Liu](#)
- [Xiaolong Ma](#)
- [Abhijit Mahalanobis](#)
- [Michael M. Marefat](#)
- [Jerzy W. Rozenblit](#)
- [Soheil Salehi](#)
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- [Hal S. Tharp](#)
- [Changlong Wu](#)
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- [Han Xu](#)
- [Huanrui Yang](#)
- [Qingzhao Zhang](#)

- [Danella Zhao](#)



Biomedical Technologies

- [Ehsan Azimi](#)
- [Ali Bilgin](#)
- [Siyang Cao](#)
- [Eungjoo Lee](#)
- [Michael M. Marefat](#)
- [Janet Meiling Roveda](#)
- [Jerzy W. Rozenblit](#)
- [Soheil Salehi](#)
- [Hao Xin](#)



Circuits, Microelectronics and Very-Large-Scale Integration

- [Kathleen Melde](#)
- [Janet Meiling Roveda](#)
- [Soheil Salehi](#)
- [Hao Xin](#)
- [Danella Zhao](#)



Communications, Coding and Information Theory

- [Boulat Bash](#)
- [Ali Bilgin](#)
- [Christos Gagatsos](#)
- [Marwan Krunz](#)
- [Michael W. Marcellin](#)
- [Kathleen Melde](#)
- [Narayanan Rengaswamy](#)
- [Ravi Tandon](#)
- [Bane Vasić](#)
- [Changlong Wu](#)
- [Hao Xin](#)



Computer Architecture and Cloud/Distributed Computing

- [Tosiron Adegbija](#)
- [Ali Akoglu](#)
- [Boulat Bash](#)
- [Tomas Cerny](#)
- [Jyotikrishna Dass](#)
- [Narayanan Rengaswamy](#)
- [Soheil Salehi](#)
- [Huanrui Yang](#)
- [Danella Zhao](#)



Optics, Photonics and Terahertz Devices and Systems

- [Ehsan Azimi](#)
- [Boulat Bash](#)
- [Christos Gagatsos](#)
- [Shelbi Jenkins](#)
- [Abhijit Mahalanobis](#)
- [Kelly Simmons-Potter](#)
- [Janet Meiling Roveda](#)
- [Hao Xin](#)



Quantum Information Science and Engineering

- [Boulat Bash](#)
- [Christos Gagatsos](#)
- [Shelbi Jenkins](#)
- [Narayanan Rengaswamy](#)
- [Bane Vasić](#)



Signal, Image and Video Processing

- [Ehsan Azimi](#)
- [Ali Bilgin](#)
- [Siyang Cao](#)
- [Jyotikrishna Dass](#)
- [Marwan Krunz](#)
- [Eungjoo Lee](#)
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- [Narayanan Rengaswamy](#)
- [Jeffrey J. Rodriguez](#)
- [Changlong Wu](#)
- [Han Xu](#)

- [Huanrui Yang](#)



Software Engineering and Embedded Systems

- [Tosiron Adegbiya](#)
- [Ehsan Azimi](#)
- [Tomas Cerny](#)
- [Sen He](#)
- [Eungjoo Lee](#)
- [Michael M. Marefat](#)
- [Jerzy W. Rozenblit](#)
- [Soheil Salehi](#)
- [Pratik Satam](#)



Wireless Networking, Security and Systems

- [Boulat Bash](#)
- [Tomas Cerny](#)
- [Christos Gagatsos](#)
- [Marwan Krunz](#)
- [Loukas Lazos](#)
- [Ming "Fred" Li](#)
- [Xiaolong Ma](#)
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- [Jerzy W. Rozenblit](#)
- [Soheil Salehi](#)
- [Pratik Satam](#)
- [Hongyi "Michael" Wu](#)
- [Hao Xin](#)
- [Han Xu](#)
- [Qingzhao Zhang](#)